POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Year : 2025
Programme: BE
Full Marks : 100
Course: Basic Electrical Engineering (New)
Pass Marks : 45
Time : 3 hrs.

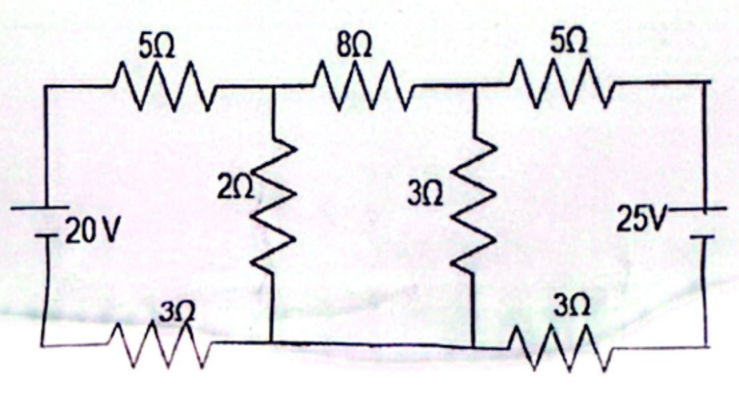
Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

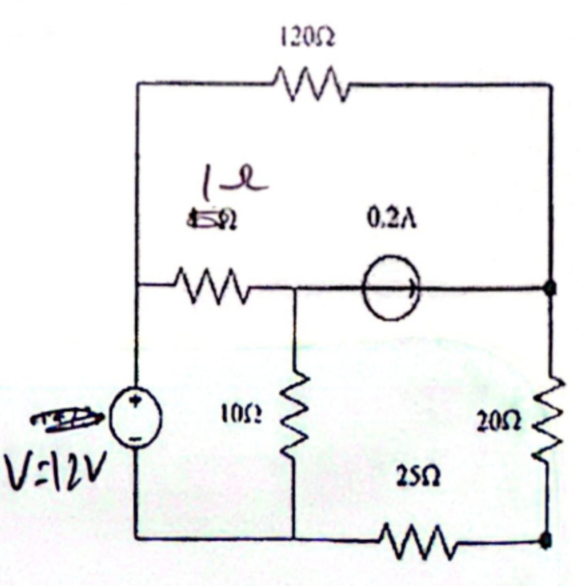
Attempt all the questions.

1. a) Describe in brief the generation, transmission, distribution and consumption of electrical energy. 7

b) Find the current across 8Ω resistor using nodal analysis.

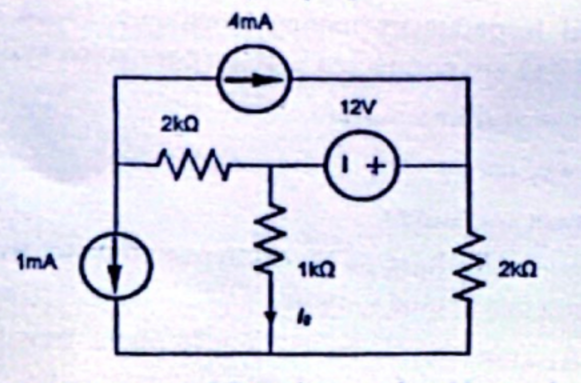


2. a) What should be the value of load resistance in place of 20 Ω such that maximum power will transferred to it. Also find the maximum power transferred to it. 8

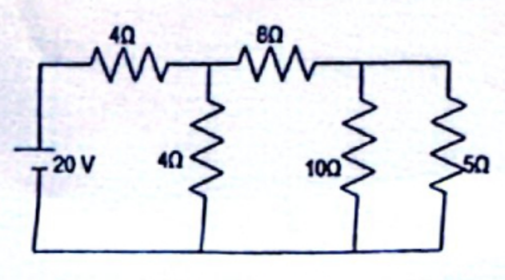


OR

Find current through 1KΩ using Superposition Theorem.



b) Calculate the current across 5Ω resistor using Norton's theorem. 7



3. a) Find the average value, effective value and form factor for a sinusoidal voltage waveform. 7

b) Two circuits A and B are connected in parallel across a 200V, 50 Hz supply mains. Circuit A consists of a resistance of 100Ω and inductance of 0.12H in series, circuit B consists of a resistance of 20Ω in series with capacitor of 40μF. Calculate: 8

- i. current in each branch
- ii. Source current
- iii. circuit power factor
- iv. Real power, reactive power and apparent power.

Also draw phasor diagram.

4. a) A series R-L-C circuit has 10Ω resistance, 0.01H inductance, and 100μF capacitor. A 100V supply with variable frequency is applied to this circuit and tuned until resonance occurs. Find the resonant frequency, bandwidth, and quality factor. current drawn by the circuit, and voltage across inductor at resonance condition. 7

OR

Derive the necessary expression for resonant frequency in RLC series circuit.

b) Three equal impedances having resistance 15 Ω and inductive resistance 12 Ω are connected in delta connected system. Find; 8

- i. Phase and line current
- ii. Power factor
- iii. Power consumed

5. a) Explain how three phase power is measured using two wattmeter in star and delta connected system. 7

b) A 20 KVA, 200/200 V, 50 Hz single phase transformer has the following test results. 8

	Volt	Ampere	Watt
Open circuit test with H.V. open	200	4	120
Short circuit test with L.V. shorted	60	10	300

Calculate the parameter of equivalent circuit of a transformer referred to H.V. side.

6. a) The power input to a 230 volts DC shunt motor is 8.477 kW. The field resistance is 230 ohm and armature resistance is 0.28 ohm. Find the input current, armature current and back e.m.f. 7

b) Explain the principle of operation of three phase induction motor. 8

7. Write short notes on: (Any two) 2×5

- a) Kirchhoff's law
- b) Power factor and its significance
- c) Transformer Losses

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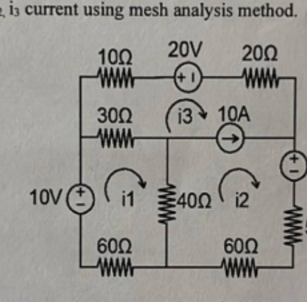
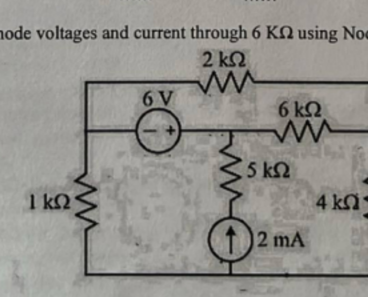
Attempt all the questions.

1. a) Explain in brief the role of electricity in modern society. Explain generation, transmission and distribution of electrical energy in context of Nepal with the help of single line diagram. 7

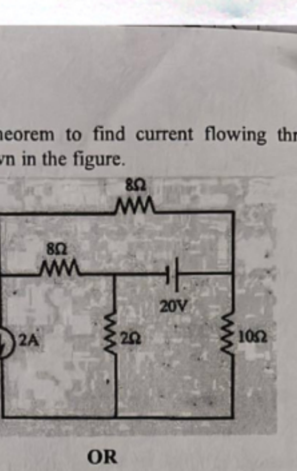
b) Find i_1, i_2, i_3 current using mesh analysis method. 8



2. a) Find node voltages and current through 6 KΩ using Nodal analysis. 7

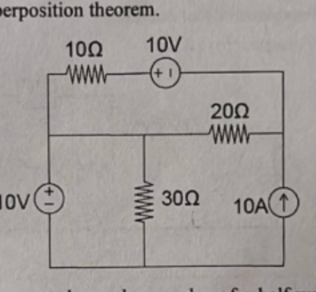


b) Use Superposition theorem to find current flowing through the 10 ohms resistor shown in the figure. 8

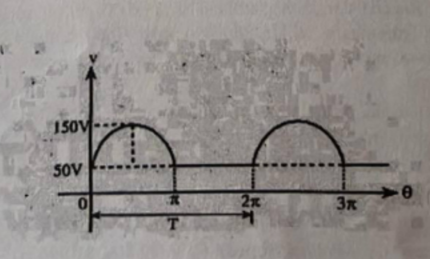


OR

State superposition theorem. Also find the voltage across 10Ω resistor using superposition theorem.

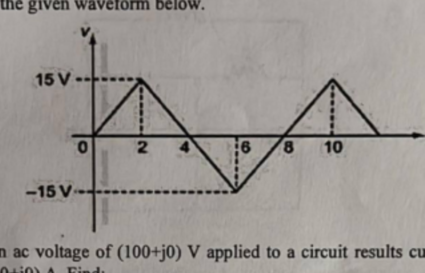


3. a) Calculate the average value and rms value of a half wave rectified sinusoidal voltage as shown in the figure. 7



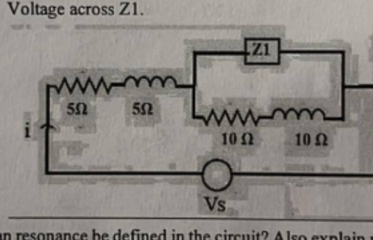
OR

Calculate the average value, RMS (effective) value and form factor of the given waveform below.



b) An ac voltage of (100+j0) V applied to a circuit results current of (10+j0) A. Find: 8

- i. Impedance Z1 of circuit.
- ii. Voltage across Z1.



4. a) How can resonance be defined in the circuit? Also explain resonance in series RLC circuit. 7

b) A balanced star connected load of (8+j6) Ω per phase is connected to a three-phase, 50 Hz, 380 V supply. Find the phase current, line current, power factor, active power, reactive power and total volt-amps. 8

5. a) Explain the differences and analogy between magnetic circuit and electric circuit. 7

b) A 50-kVA, 2400V/120V, 50-Hz, single-phase transformer gave the following test results: 8

OC test (HV winding open): 120 V, 9.65 A, 396 W.

SC test (LV winding shorted): 92 V, 20.8 A, 810 W.

Find:

- i. the equivalent circuit parameters as referred to LV side.
- ii. the efficiency at half load,0.8 pf lagging.

6. a) A 220 V dc shunt motor with an armature resistance 0.5 Ω runs at 600 rpm on full load and takes an armature current of 20A. If a resistance of 1Ω is placed in the armature circuit, find the speed at i) full load torque ii) half load torque 8

b) Explain the speed control methods of DC shunt motor with suitable equations and characteristic curves. 7

7. Write short notes on: (Any two) 2×5

- a) KVL and KCL
- b) Power Factor and its significance
- c) Condition for maximum efficiency of transformer

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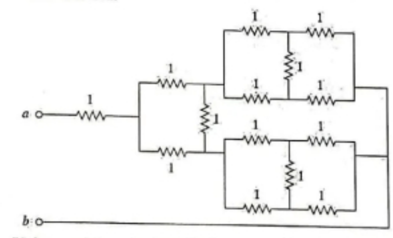
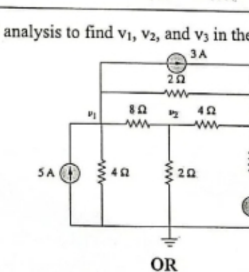
Attempt all the questions.

1. a) Classify types of consumers based upon their electricity consumption pattern. Also draw the single line diagram of a general power system. 7

b) Find equivalent resistance at a-b R_{ab} in the figure. Assume each element is 1Ω. 8



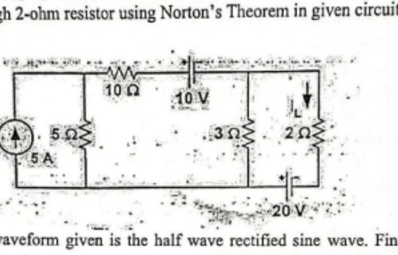
2. a) Using nodal analysis to find v_1, v_2, and v_3 in the following circuit. 7



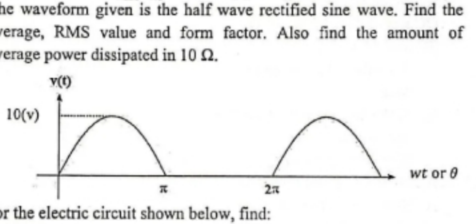
OR

What are active and passive elements, unilateral and bilateral element? What is star delta transformation and writes its role in circuit analysis?

b) State Superposition Theorem power theorem. Find voltage drop through 2-ohm resistor using Norton's Theorem in given circuit. 8

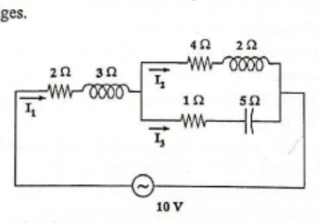


3. a) The waveform given is the half wave rectified sine wave. Find the average, RMS value and form factor. Also find the amount of average power dissipated in 10 Ω. 8



b) For the electric circuit shown below, find: 7

- i. supply current and branch currents
- ii. voltage across each branch
- iii. Active, reactive and apparent power of entire circuit
- iv. overall power factor
- v. Draw phasor diagram showing the currents and supply voltages.



4. a) With the help of series resonance curve, explain resonance in series RLC circuit. 7

b) Three identical coils, each of resistance 10 Ω and inductance 42 mH are connected a) in star b) in delta to a 415 V, 50 Hz, 3-φ supply. Determine: 8

- i. line and phase currents
- ii. Line and phase voltages
- iii. the total power dissipated in each case and compare them.

5. a) Explain two wattmeter method for three phase power measurement. Also, how it is different from one wattmeter and three wattmeter method. 7

b) A 5 kVA, 200/1000 V, 50 Hz, single phase transformer gave the following test results: 8

OC test: 200V, 1.2 A, 90 W

SC test: 50 V, 5A, 110 W

- i. Calculate the parameters of the equivalent circuit referred to LV side.
- ii. Calculate the output secondary voltage when delivering 3 kW at 0.8 power factor lagging, the input voltage being 200 V. Find also the percentage voltage regulation.

6. a) A 250 V DC shunt motor takes 8 A line current on no load and runs at 1000 rpm. The resistance of the field winding and armature winding are 200 Ohm and 0.8 Ohm respectively. If the full load line current is 20 A, calculate the full load speed assuming constant air gap flux. 7

b) Explain the construction of DC generator with working principle. 8

OR

Explain the importance of transformer in power transmission. Derive the EMF equation of transformer.

7. Write short notes on: (Any two) 2×5

- a) Speed Control of DC Motor
- b) Color coding of resistors
- c) Power in purely inductive circuit

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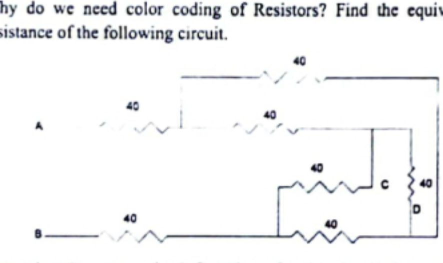
Attempt all the questions.

1. a) With the help of single line diagram, explain generation, transmission, and distribution of electricity in Nepal. 7

OR

Explain the present energy scenario in Nepal.

b) Why do we need color coding of Resistors? Find the equivalent resistance of the following circuit. 8



2. a) Determine the current in 5 Ω resistor for the circuit shown using Superposition Theorem (Figure 22). 8

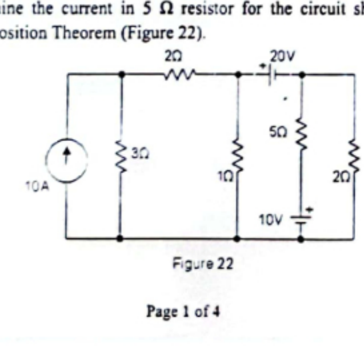
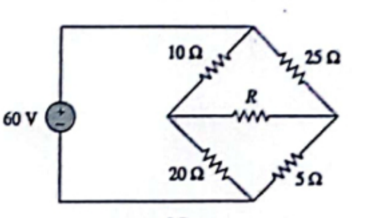


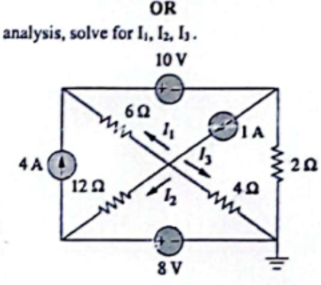
Figure 22

b) Calculate the value of R that will absorb maximum power from the circuit (shown in the figure). Also calculate the maximum power drawn by it. 7



OR

Using mesh analysis, solve for I_1, I_2, I_3.



3. a) Determine the V_1 and V_2 using Nodal Analysis method. Also determine the current flowing through 30 Ω resistor (Figure 19) 7

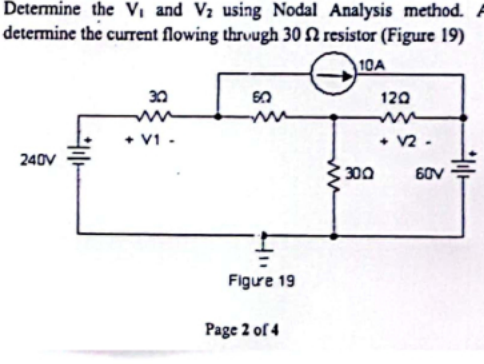
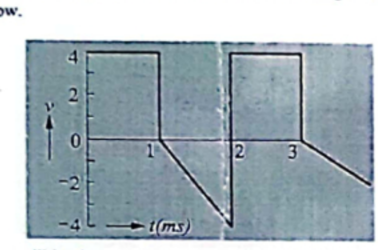


Figure 19

b) Find the average value, rms value of the voltage waveform given below. 8



4. a) What will be the case for resonance in series RLC circuit? Derive the resonant frequency for this case. 7

b) A circuit have following electrical components in parallel: a) a resistor of 500 ohm, b) an inductance of 2H, c) capacitor of 10 microfarad. In this circuit 200V, 50 Hz source is applied. Calculate: 8

- i. the total current drawn from the supply
- ii. complex power
- iii. active power
- iv. reactive power
- v. power factor of the circuit

5. a) The star connected stator of three phase, 50 Hz alternator supplies a balanced delta connected load. Each phase of the load consists of a coil of resistance of 15Ω and inductance 36mH and the phase voltage generated by the alternator is 231V. Calculate: 8

- i. the phase and line currents
- ii. the load power factor
- iii. the power delivered to the load.

b) An ideal 25 KVA transformer has 500 turns on the primary winding and 40 turns on the secondary winding. The primary winding is connected to a 3.3 KV supply. Calculate: 7

- i. The secondary emf.
- ii. The primary and secondary currents on full load.
- iii. The maximum core flux.

6. a) Explain the operation of transformer on load and also draw phasor diagram at lagging power factor. 7

b) A 220 Volts dc shunt motor draws a current of 5 A. and runs with a speed of 1500 rpm at no-load. When the motor is loaded, the speed drops to 1400 rpm. Calculate the current drawn by the motor at loaded condition. Given that the armature winding resistance is 0.2 Ω and field winding resistance is 110 Ω. 8

7. Write short notes on: (Any two) 2×5

- a) Voltage divider Rule
- b) Sources of Energy
- c) Operation of DC motors

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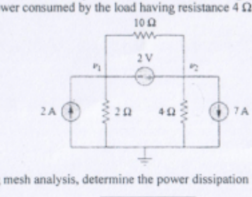
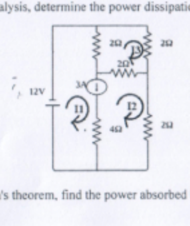
Attempt all the questions.

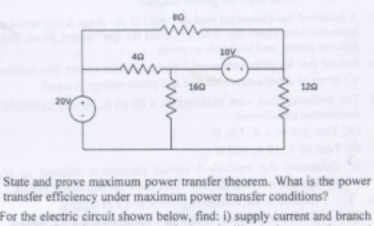
1. a) What are nodes and loop in electric circuit? For the given electric circuit, use nodal analysis to find the current through, voltage across and power consumed by the load having resistance 4 Ω. 7



b) Using mesh analysis, determine the power dissipation in 4 Ω resistor. 8

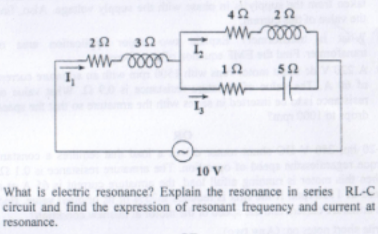


2. a) Using Thevenin's theorem, find the power absorbed by 12Ω resistor. 8



b) State and prove maximum power transfer theorem. What is the power transfer efficiency under maximum power transfer conditions? 7

3. a) For the electric circuit shown below, find: i) supply current and branch currents ii) voltage across each branch iii) Active, reactive and apparent power of entire circuit iv) overall power factor v) Draw phasor diagram showing the currents and supply voltages. 7



b) What is electric resonance? Explain the resonance in series RL-C circuit and find the expression of resonant frequency and current at resonance. 8

OR

A voltage v(t) = 100 sin 314t is applied to a series circuit consisting of 10 Ω resistance, 0.0318 H inductance, and a capacitor of 63.6 μF. Calculate the followings:

- i. Expression for i(t)
- ii. Power factor

- iii. Active, reactive, and apparent power
- iv. Also draw the phasor diagram.

4. a) A balanced star-connected load of (8+j6) Ω per phase is connected to a balanced three-phase 400 V supply. Find the line current, power factor, reactive power, and total volt-amperes. 8

b) Proved that for delta connected 3 phase balance system line current is √3 times of phase current but line and phase voltage is equal. 7

5. a) The following data were obtained on a 20 kVA, 50 Hz, 2,000/200 V distribution transformer. 10

OC Test: 200 V, 4 A, 120 W

SC Test: 60 V, 10 A, 300 W

- i) Determine the equivalent circuit parameters referred to high voltage and low voltage side. Also, draw the equivalent circuit.
- ii) Determine the voltage regulation at 0.8 pf lagging.
- iii) Calculate efficiency at full load condition at 0.8 p.f. lagging

b) An inductive coil of resistance 10 Ohm and inductance 0.1 H is connected in parallel with a 150 μF capacitor to a variable frequency, 200 V supply. Find the resonant frequency at which the total current taken from the supply is in phase with the supply voltage. Also, find the value of this current. 5

6. a) What is transformer? Explain two major application area of transformer. Find the EMF equation of transformer. 7

b) A 220 V dc shunt motor runs with 1500 rpm with an armature current of 60 A. The value of armature resistance is 0.9 Ω. What value of resistance is to be inserted in series with the armature so that the speed drops to 1000 rpm? 8

OR

A 20 hp, 250 V DC shunt motor drives a load that requires a constant torque regardlessthe speed of operation. The armature resistance is 0.1 Ω. When this motor is running atfull load, the armature current is 65 A at a speed of 1100 rpm. If the flux is reduced to75% of its original value, find the armature current and the speed of the motor at thisnew condition?

7. Write short notes on: (Any two) 2×5

- a) Generation, transmission and distribution of electricity using single line diagram
- b) Working principle and uses of three phase induction motor
- c) Power factor and its significance